

Financial Blind Spot Detection Using Transaction Behaviour Analytics and Machine Learning

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Abstract

Decision-makers today receive more data, dashboards, and metrics than ever before. While analytics aims to support better decisions, excessive or poorly structured information often creates confusion, hesitation, and reduced confidence an experience known as **analytics anxiety**. This paper explains how data overload affects the cognitive and emotional processes of decision-makers. Drawing from cognitive load theory, information overload research, and decision science, we identify three main culprits: metric overload (those dashboards with 20+ KPIs staring at you), unclear prioritization (which number should I even look at first?), and missing context (why did this metric change, and what should I do about it?). Through a systematic review spanning cognitive psychology, HCI research, and recent industry reports (Gartner 2023; Deloitte 2022), we develop the **ACE Model**—a three-layer framework addressing the psychological roots of analytics anxiety, the design flaws that trigger it, and evidence-based solutions. Our centerpiece is the **5-3-1 Design Principle**: show 5 core metrics maximum, highlight 3 key insights, recommend 1 clear action. Simple? Yes. Easy to implement? That's the point. This framework sits at the intersection of data science, cognitive psychology, and human-computer interaction, offering practical guidelines for analytics tools that actually help rather than overwhelm.

Keywords: Financial Blind-Spot Detection, Transaction Behaviour Analytics, Predictive Modelling (Random Forest), SHAP Explainability, Synthetic Stress Testing

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